

CSE422 Lab Project: Loan Status Prediction

Exploratory Data Analysis and Machine Learning Classification

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Dhaka, Bangladesh Course: CSE 422 *Section: 14*

Abstract—This report details the implementation of a comprehensive machine learning pipeline designed to predict loan approval outcomes using a financial dataset of 45,000 records. The methodology encompasses exploratory data analysis, dataset preprocessing, and the comparative evaluation of multiple classification models, including a Neural Network (MLP), Decision Tree, and Logistic Regression, alongside unsupervised K-Means clustering. The Neural Network demonstrated superior predictive capabilities, achieving the highest classification accuracy of 91.53% and an Area Under the Curve (AUC) of 0.9660, making it the most robust model for this financial risk assessment task.

I. INTRODUCTION

Credit risk assessment is a foundational challenge in the financial sector. Accurately predicting whether an applicant will default on a loan or successfully repay it allows institutions to minimize financial loss and automate the screening process. The aim of this project is to build a robust predictive model that maps a user's personal and financial background to their loan eligibility.

For this project, the chosen target feature is `loan_status`. This is a discrete, binary feature where a value of 1 signifies a granted/successful loan, and a value of 0 indicates a denied loan or a default. By analyzing the dataset and treating it as a classification problem, the objective is to uncover patterns that strongly correlate with high-risk applicants.

II. DATASET DESCRIPTION

The predictive models were trained and evaluated using the provided `14.csv` dataset. The structural profile of this dataset is as follows:

- **Data Volume:** The dataset contains 45,000 individual records (data points), representing a substantial sample size for training generalized machine learning models.
- **Dimensionality:** There are 14 distinct features (columns) for each record.
- **Feature Types:** The dataset consists of a heterogeneous mix of variables. It includes continuous numerical data (e.g., `person_income`, `loan_amnt`, `loan_int_rate`), discrete numerical data (e.g., `person_age`), and categorical textual data (e.g., `person_gender`, `person_education`, `person_home_ownership`).

III. DATA ANALYSIS

Before applying machine learning algorithms, Exploratory Data Analysis (EDA) was conducted to observe the underlying distributions and relationships within the data. A correlation matrix was generated for all numerical features to measure linear dependencies.

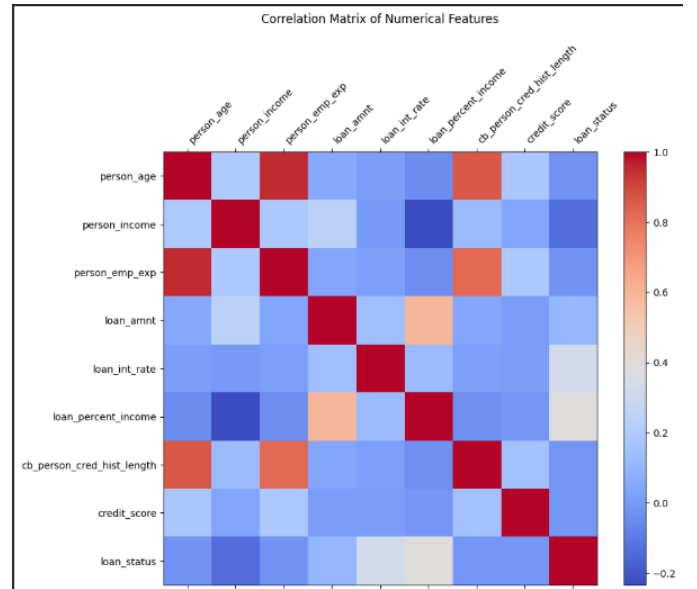


Fig. 1. Correlation matrix of numerical features, visualizing the magnitude of linear relationships between financial attributes.

The heatmap (Fig. 1) allows for the identification of multicollinearity. For instance, analyzing how variables like `person_income` or `credit_score` correlate with the `loan_amnt` can provide early indicators of which features will heavily influence the model's decision-making process.

IV. DATASET PRE-PROCESSING

Raw data is rarely suitable for direct ingestion into machine learning algorithms. A robust pre-processing pipeline was executed to resolve inconsistencies and format the data appropriately.

- **Handling Null and Duplicate Values:** Missing data can cause mathematical operations in algorithms to fail.

The dataset was scanned for null values and duplicate rows. The analysis confirmed 0 missing values and 0 duplicates, ensuring high data integrity without the need for statistical imputation.

- **Encoding Categorical Values:** Machine learning models compute distances and probabilities using mathematics, which means text data like “Bachelor” or “Master” must be converted into numerical formats. Scikit-learn’s `LabelEncoder` was applied to assign a unique integer to every category within the text-based columns.
- **Feature Scaling:** The dataset features operate on drastically different scales (e.g., `person_age` ranges in the tens, while `person_income` ranges in the tens of thousands). Without scaling, models like Neural Networks or distance-based algorithms heavily bias toward features with larger magnitudes. `StandardScaler` was utilized to standardize features by removing the mean and scaling to unit variance, ensuring all variables contributed equally to the model’s weight updates.

V. MODEL TRAINING

The pre-processed data was split using an 80/20 ratio, allocating 80% of the data for training and holding out 20% for unbiased testing. The project applied both supervised and unsupervised learning techniques:

- **Neural Network:** A Multi-Layer Perceptron (`MLPClassifier`) was deployed with two hidden layers containing 32 and 16 neurons, respectively. This architecture allows the model to map complex, non-linear representations of the data.
- **Baseline Supervised Models:** To establish a comparative baseline, a Logistic Regression model (to test for linear separability) and a Decision Tree Classifier (to test hierarchical rule-based splits) were trained.
- **Unsupervised Learning:** The target column (`loan_status`) was temporarily removed to treat the dataset as an unsupervised problem. K-Means clustering ($k = 2$) was applied to determine if the financial attributes naturally segregate into distinct high-risk and low-risk groupings without explicit labels.

VI. MODEL EVALUATION

The performance of the three supervised classifiers was evaluated using the testing subset. The quantitative metrics are summarized in Table I.

TABLE I
MODEL COMPARISON METRICS

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Reg.	0.8901	0.7663	0.7308	0.7482	0.9483
Decision Tree	0.8984	0.7740	0.7701	0.7721	0.8527
Neural Net (MLP)	0.9153	0.8728	0.7269	0.7932	0.9660

The confusion matrices shown in Fig. 2 provide a detailed view of the classification performance of the three models. The Neural Network produced 6,777 correctly classified denied-loan cases and 1,461 correctly classified approved-loan cases.

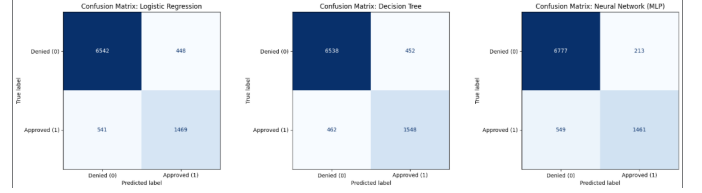


Fig. 2. Confusion Matrices comparing the classification results of Logistic Regression, Decision Tree, and Neural Network models on the test set.

It also produced 213 false positives and 549 false negatives. Compared with the other models, the Neural Network achieved the lowest number of false positives, demonstrating its ability to reduce incorrect approval predictions.

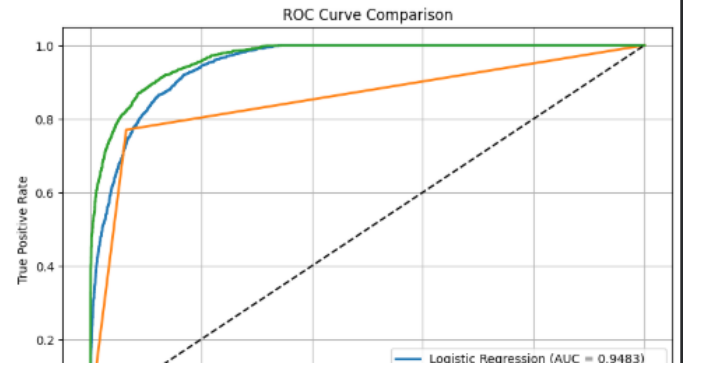


Fig. 3. ROC Curve comparison highlighting the AUC scores of the trained models.

Furthermore, the Receiver Operating Characteristic (ROC) curves (Fig. 3) confirm the Neural Network’s superiority. The MLP achieved the highest Area Under the Curve (AUC) of 0.9660, indicating that it has the strongest capability to distinguish between the positive and negative classes across various probability thresholds.

VII. CONCLUSION

Based on the comprehensive evaluation matrix, the **Neural Network (MLP)** emerged as the best-performing model. It secured the highest accuracy (91.53%), the highest F1 Score (0.7932), and the most robust AUC (0.9660).

The probable reason behind the Neural Network’s success lies in the complexity of financial data. Attributes such as age, income, and credit history rarely share a strict linear relationship with loan defaults. While Logistic Regression struggled due to linear constraints and the Decision Tree likely suffered from overfitting or rigid splits, the Neural Network’s hidden layers successfully captured the complex, multi-dimensional, and non-linear interactions between these financial indicators, resulting in highly accurate predictions.